Vaishali Sonker

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EDUCATION

Bachelor of Computer Science and Engineering, CSJMU Kanpur 7.9 CGPA 2023- 2027

SKILLS

Technical Skills

- **Programming Languages:** Python, C++, SQL
- **Machine Learning:** Scikit-learn, TensorFlow (basics), NumPy, Pandas, Matplotlib, Seaborn
- **Deep Learning:** Neural Networks, CNN, ANN (basics)
- **NLP Tools:** NLTK, TF-IDF, Text Preprocessing
- **Web Frameworks:** Streamlit, Flask (basics)
- **Data Science:** Data Cleaning, EDA, Feature Engineering, Model Evaluation
- **Version Control:** Git, GitHub
- **Tools:** Jupyter Notebook, VS Code, Google Colab

Core Competencies

- Data Structures & Algorithms (DSA)
- Problem Solving & Debugging
- Data Visualization
- Model Building & Evaluation
- Team Collaboration (Hackathons & Projects)
- Presentation & Report Writing

PROJECTS

1. Heart Disease Prediction System

- Tech Stack: Python, Scikit-learn, Pandas, Streamlit, Joblib
- Description:
  Developed a machine learning model using K-Nearest Neighbors (KNN) and other classification algorithms to predict the likelihood of heart disease based on patient health data.
- Key Features:
  - Data preprocessing and feature scaling
  - Model comparison based on accuracy and precision
  - Deployed an interactive Streamlit web app for real-time prediction
- Outcome: Achieved high accuracy and built a fully functional web interface for users.

2. Spam Mail Detection System

- Tech Stack: Python, Scikit-learn, NLTK, Pandas, Streamlit
- Description:
  Built a text classification model to detect spam and ham emails using Natural Language Processing (NLP).
- Key Features:
  - Data cleaning and preprocessing with tokenization and stopword removal
  - TF-IDF feature extraction
  - Used Naive Bayes and Logistic Regression models for classification
  - Deployed a simple UI for text input and prediction
- Outcome: Accurate detection of spam messages using NLP pipelines.

3. Astronomical Instrument Management System (SIH Project)

- Tech Stack: Python, Flask/Streamlit, MySQL (proposed), API integration
- Description:
  Proposed an innovative solution for cataloging and analyzing Indian astronomical instruments with data visualization and AI-based insights.
- Innovation:
  - Interactive data dashboard for researchers
  - Inclusion of ML-based classification for instrument data
  - Focused on heritage preservation and research access.
- Outcome: Submitted as part of Smart India Hackathon proposal.